Technische Universität Dresden

DRESDEN concept

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Neuroscientific Perspectives on Personality

The problem of weak associations between personality and neurocognitive measures

Weak associations between personality and neurocognitive measures
Outline

Introduction

— Problem outline

Technical issues

— Reliability

— Range restriction

Conceptual issues

— Brunswik asymmetry

— Strong vs. weak situations

Integration

Weak associations between personality and neurocognitive measures
Introduction: Problem outline

Problem outline

— **What is a weak association in the first place?**

Cohens convention does not pass the empirical test for individual differences research (and many other psychology subdisciplines!), most effects seem to be around r ~ .20-.30.

— **Then why do others observe strong associations so often?**

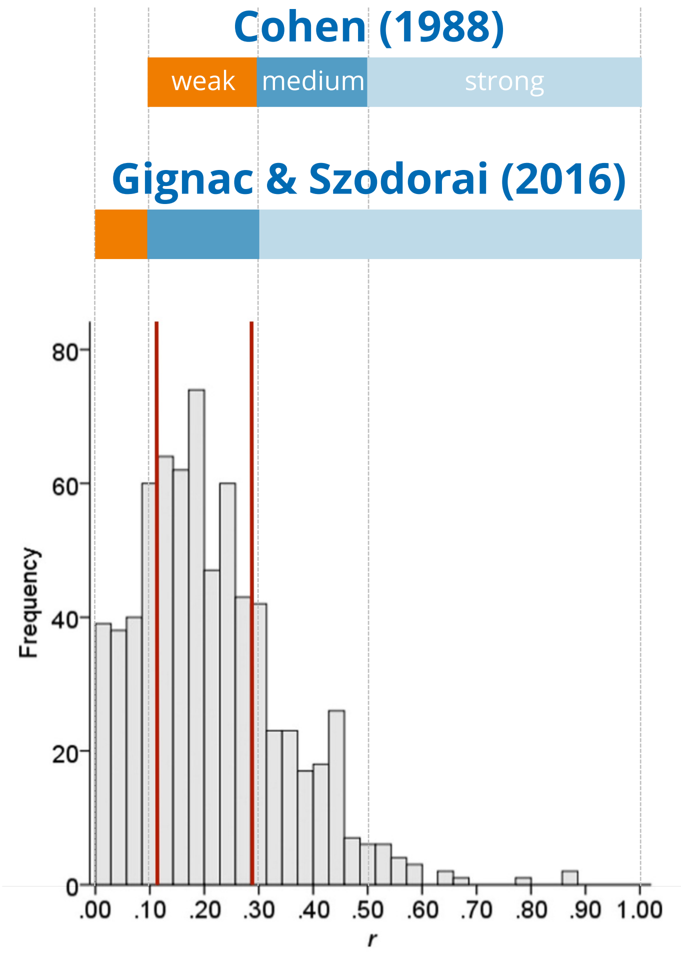
Biopsychological research is not likely to come up with stronger effects, but its effects *need to be strong* to show up in small samples (and to get published)

— **But I don't find even any significant medium associations?**

Power is maybe too low (required sample sizes to detect effects of r = .10/.20/.30 associations are N = 780/193/85 at a power of 80%)

— **But even then, my effects are smaller as to be expected?**

This may be a consequence of publication bias (see Open Science Collaboration, 2005), but also simply due to technical issues …



Gignac, G. E. & Szodorai, E. T. (2016). *PAID, 102*, 74-78.
http://doi.org/10.1016/j.paid.2016.06.069

Weak associations between personality and neurocognitive measures
Technical issues: Reliability

Correlation attenuation

— the observed correlation between two variables is weaker than their "true" relationship due to unreliability in the data or measurement error

Suboptimal reliability is a pervasive issue

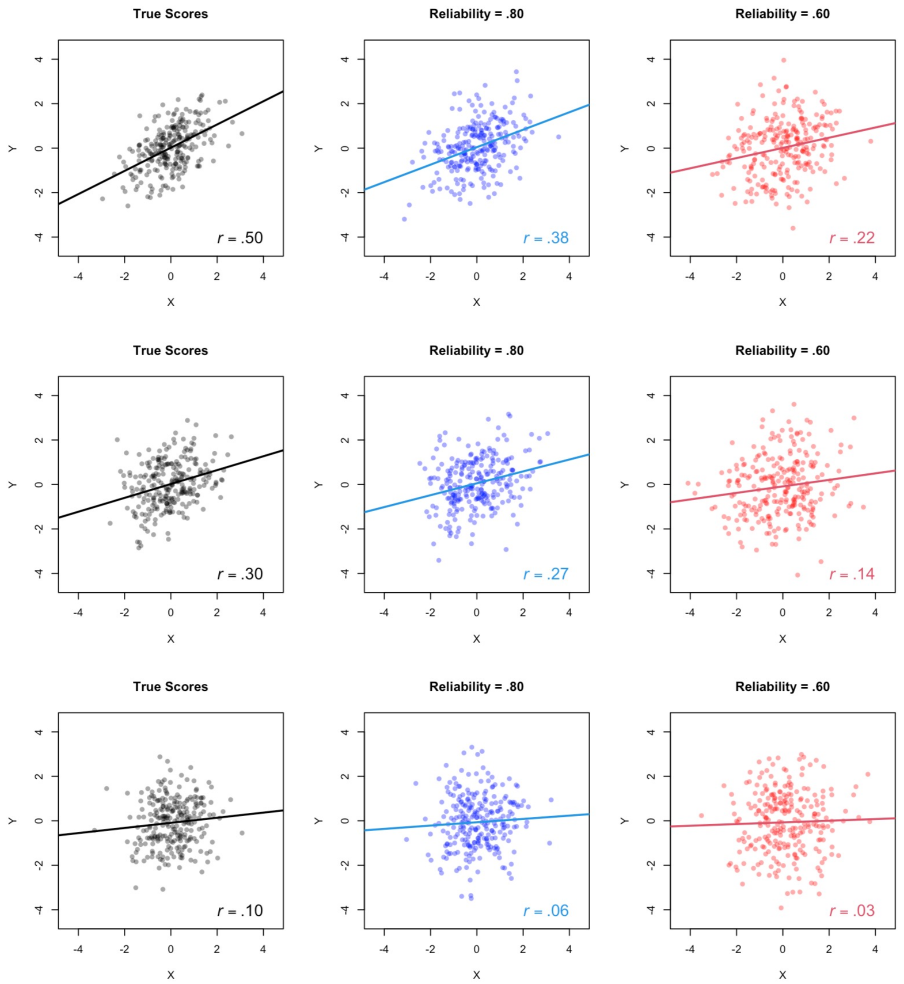
— Intelligence research: ~ .90 (own estimate)
— Personality research: ≤ .80 (own estimate)
— Cognitive research: ~ .60 (Enkavi et al., 2019)
— Imaging research: ~ .40 (Elliott et al., 2020)

Consequences

— use high reliability measures / capture measurement error

— correct expected effect size for reliability of your measures for estimation of required sample size at a given α and 1-β

— correct observed r for attenuation via $r_{true} = r_{obs}/\mathrm{sqrt}(Rel_X * Rel_Y)$



data simulated with R

Enkavi, A. Z. et al. (2019). *PNAS, 116*(12), 5472-5477.
https://doi.org/10.1073/pnas.1818430116

Elliott, M. L. et al. (2020). *Psych Sci, 31*(7), 792–806.
https://doi.org/10.1177/095679762091678

Weak associations between personality and neurocognitive measures
Technical issues: Range restriction

Correlation attenuation

— the observed correlation between two variables is weaker than their "true" relationship due to restriction of variance in the data

Range restriction is an issue in psychological research

— Participants often young and intellectually capable (and *WEIRD*)

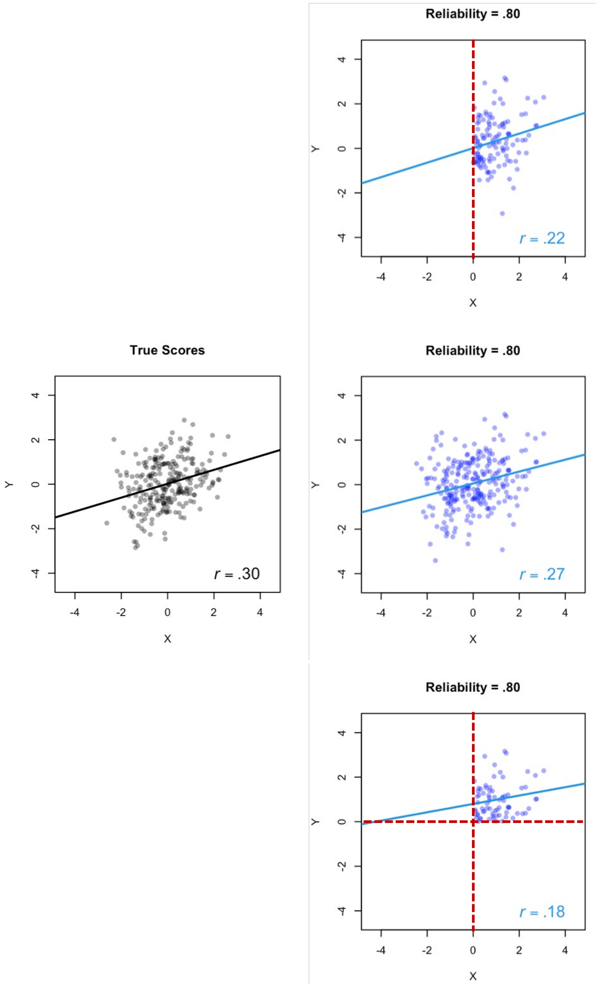
— Fictitious example: r_{true} = .30 between age and working memory is r_{obs} = .27 due to reliability issues, and r_{obs} = .22/.18 due to range restriction (only young/young and capable individuals)

Consequences

— examine individuals from general population (**not** psychology plus architecture and engineering students ;-)

— if variances in general population are known, correct observed *r* for range restriction (easiest case, variance restriction only in *a*)

$$R_{ay} = r_{aY}\left(\frac{SD_a}{sd_a}\right) / \sqrt{(1 - r_{ay}^2) + r_{ay}^2\left(\frac{SD_a^2}{sd_a^2}\right)}$$



restricted by X

unrestricted

restricted by X & Y

data simulated with R

Details and further correction methods: Carretta, T. R. & Ree, M. J. (2022). *Milit Psychol, 34*(5), 551–569.
https://doi.org/10.1080/08995605.2021.2022067

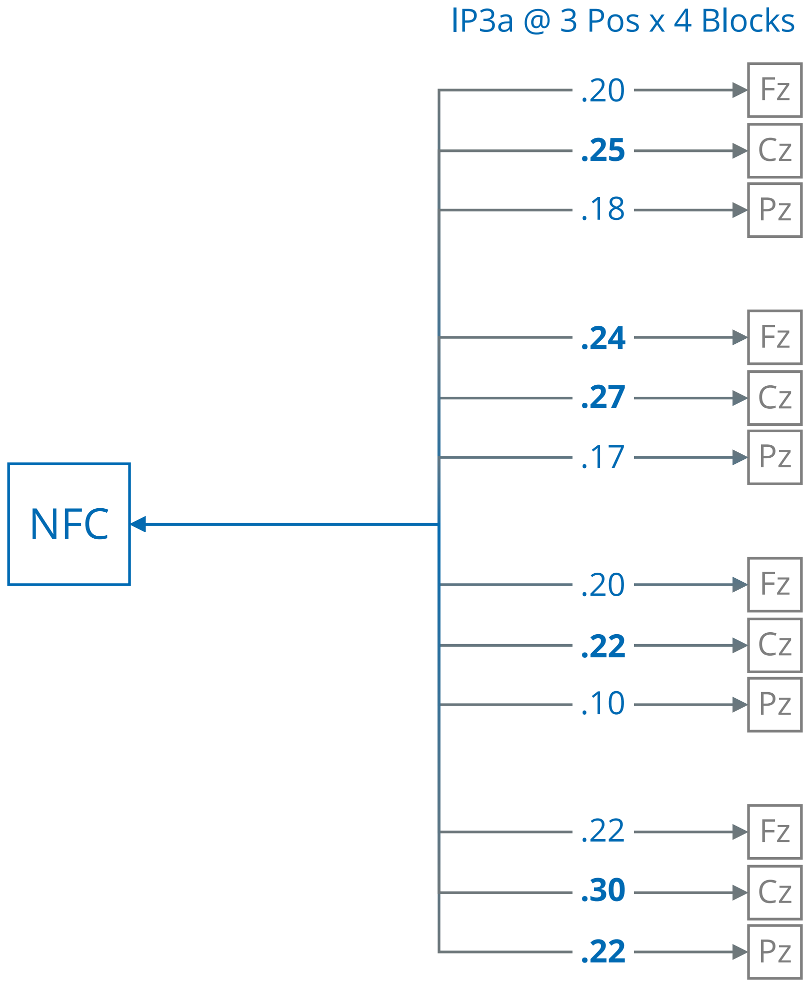
Weak associations between personality and neurocognitive measures
Conceptual issues: Brunswik asymmetry

Brunswik asymmetry (in terms of generality)

— correlations between a trait and a *specific* behavior are often smaller than those between the trait and more *general* behavior

Generality asymmetry is often a problem in psychology

— example here: we try to link a rather broad trait like Need for Cognition (NFC) to very specific EEG measures in a novelty oddball task (here: the late P3a in some block at some electrode)



unpublished results from an investigation into the relation of NFC and the Novelty lP3a, see Enge, S. et al (2008). *PSPB, 34*(6), 862-874. https://doi.org/10.1177/0146167208315556

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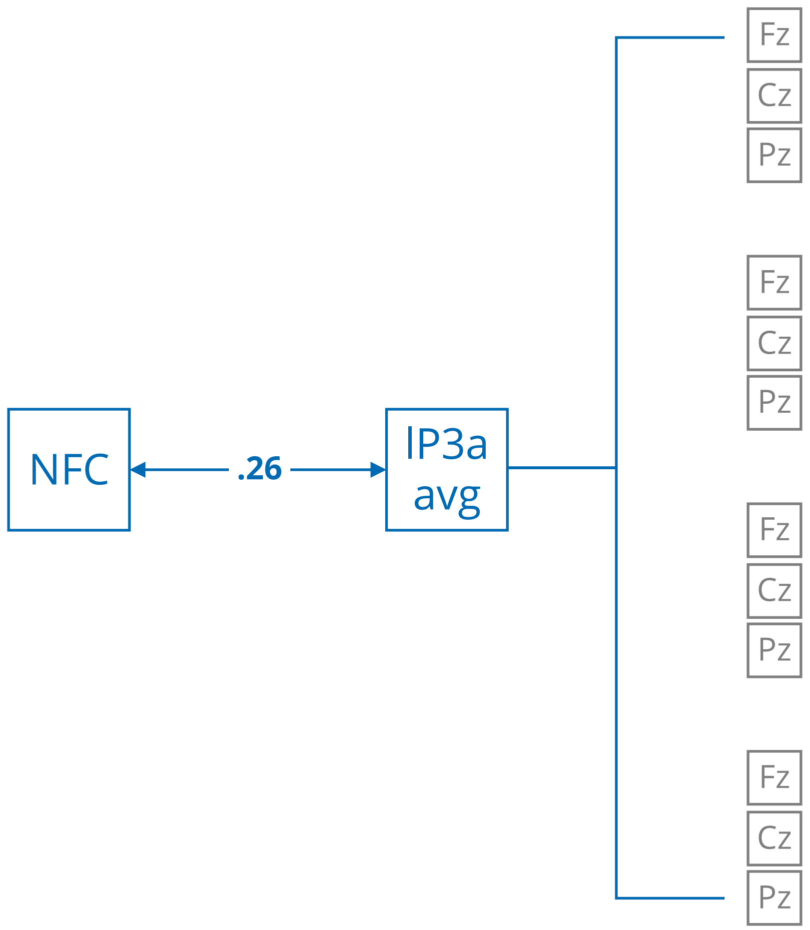
— correlations usually smaller for more asymmetrical associations



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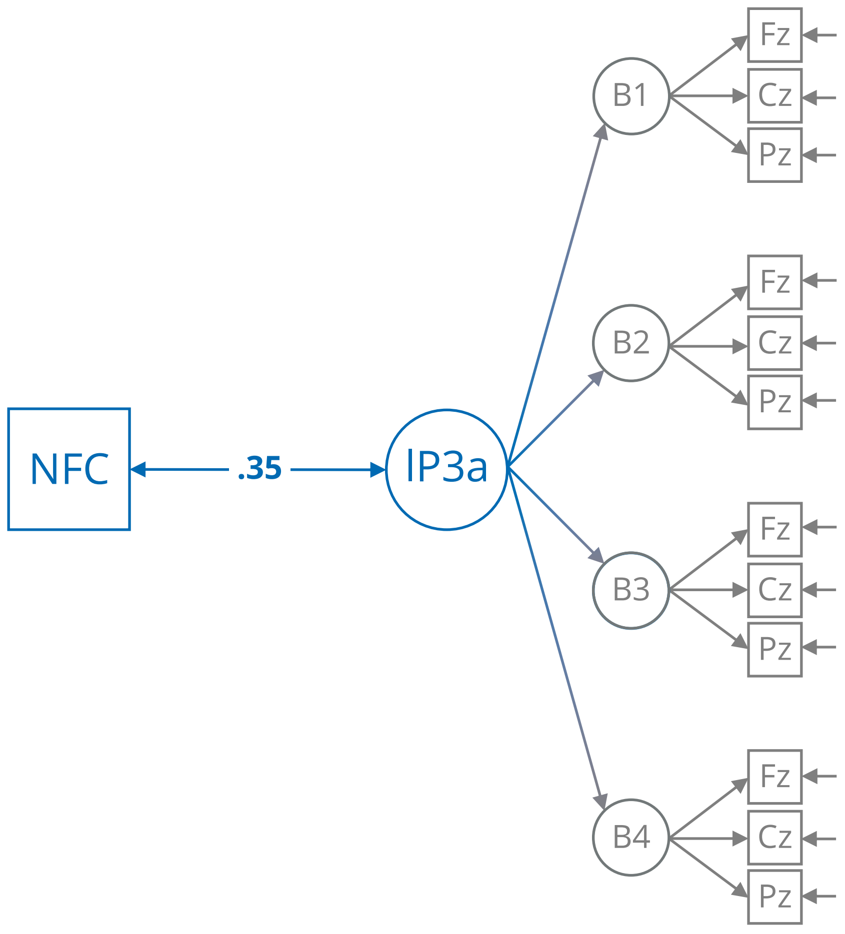
Consequences

— try to associate constructs rather than specific measures via aggregation/structural equation modelling (and by doing so, also get rid of measurement error, see above)

— considerably more effort when using more than one measure of IV and DV; arriving at a proper SEM can be a p*** in the a**



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Weak associations between personality and neurocognitive measures
Conceptual issues: Strong vs. weak situations

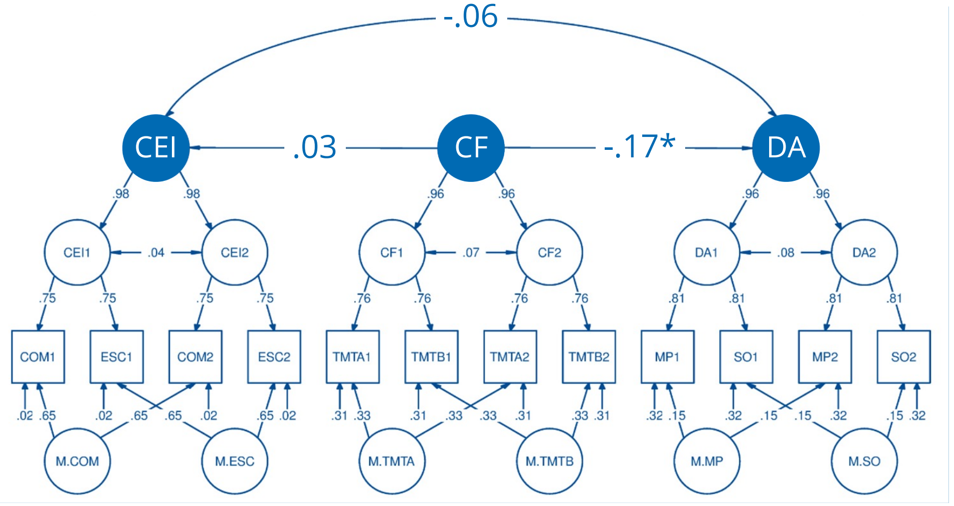
Strong vs. weak situations

— cf. Mischel: strong situations minimize, weak situations maximize individual differences

Situational affordance is an issue in cognitive studies

— experimental paradigms often *designed* to minimize the influence of individual differences in order to maximize situational influence

— individuals with different trait levels tend to behave similar due to situation strength, resulting in weak associations



Strobel et al. (2020): in a sample of *N* = 217 university students, trait *Cognitive Effort Investment*, CEI, was not related to *Demand Avoidance*, DA (controlled for cognitive functioning, CF) likely because the task was optimized to demonstrate that DA is a general phenomenon

Strobel, A., … Kührt, C. (2020). *PLoS ONE, 15*(10), e0239817.
https://doi.org/10.1371/journal.pone.0239817

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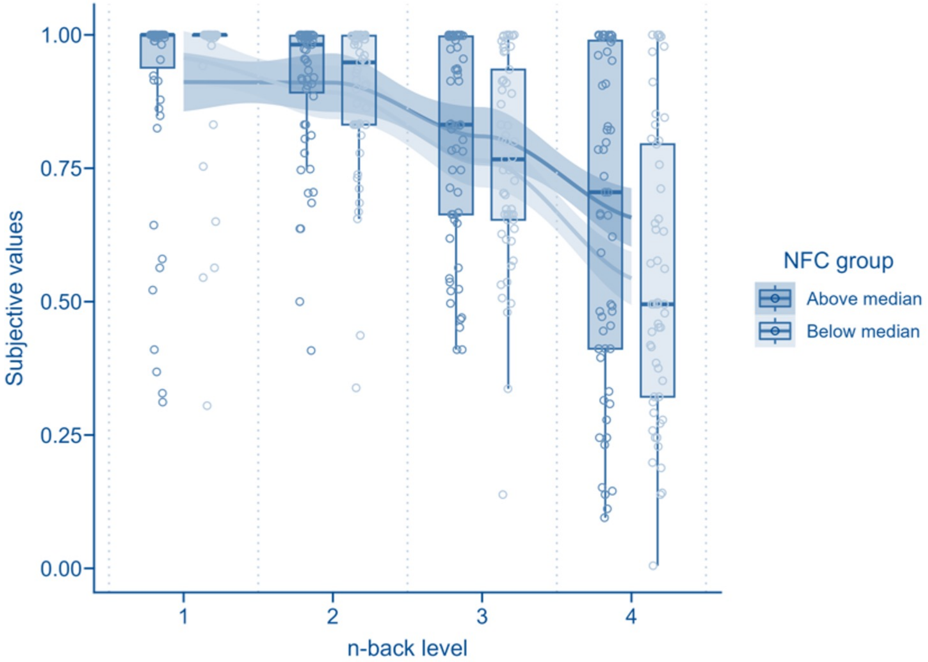
— if situation strength is reduced like in the *n*-back task and individuals are asked for subjective values of different *n*-back levels, the expected nonlinear person × situation interaction associations shows up

— tasks like the Novelty Oddball or measurements under resting conditions also create rather weak situations



Zerna, Scheffel et al. (2023): in a sample of *N* = 116 university students, high NFC individuals showed the expected pattern of subjective values over *n*-back levels

Zerna, J., Scheffel, C. et al. (2023). *SciRep*, *13*, 19501.
https://doi.org/10.1038/s41598-023-44349-3

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Consequences

— establish experimental or ecological situations with medium situation strength so that individuals can behave in line with their dispositions

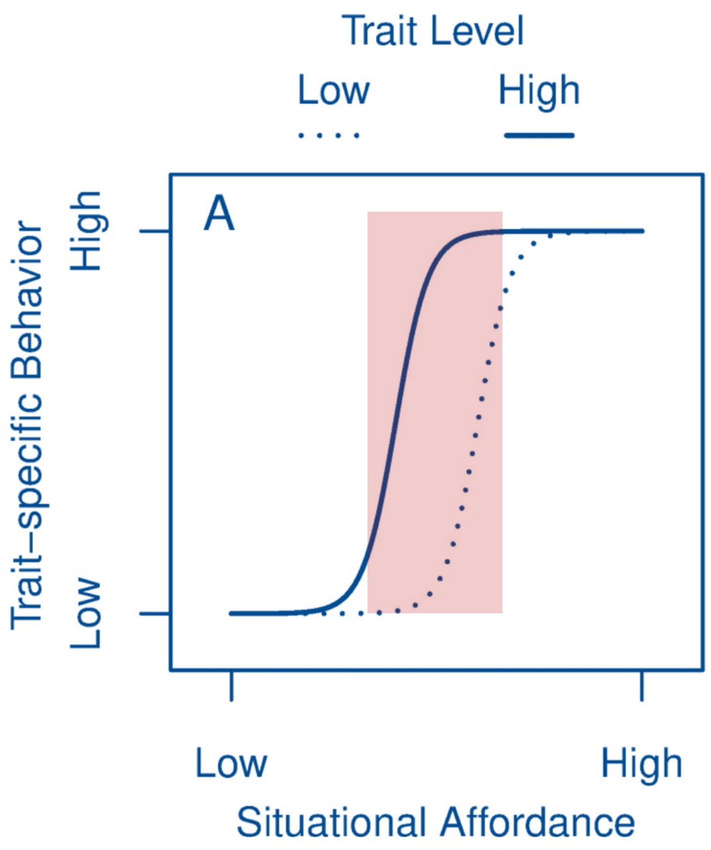

Fig. adapted from Strobel, A., … Kührt, C. (2020). *PLoS ONE, 15*(10), e0239817.
https://doi.org/10.1371/journal.pone.0239817

Weak associations between personality and neurocognitive measures

Integration: Summary and suggestions

Problem

in individual differences research, *medium* associations can be assumed at best, but they may appear *weak* because of …

Unreliability of the measures used

— carefully select measures and/or measurement models to avoid and/or correct for attenuation

Variance restriction of the samples examined

— carefully select samples and/or know true variances of your measures to correct for restriction

Generality asymmetry of the measures employed

— make sure that X and Y are of the same generality level (aggregation/SEM, also captures error)

Strong vs. weak measurement situations

— establish experimental or ecological situations with medium situation strength so that individuals can behave in line with their dispositions

Suggested reading: DeYoung, C. G., Hilger, K., et al. (2025). ***J Cogn Neurosci, 37*****(6), 1023-1034.**
https://doi.org/10.1162/jocn_a_02297

Thank you!

Details?
https://github.com/alex-strobel/PuG-2026-Talk

Questions?
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